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EXPT NO: 8

DATE: _____

**A PYTHON PROGRAM TO IMPLEMENT THE KUHN-TUCKER CONDITIONS –
LAGRANGIAN MULTIPLIER METHOD**

AIM:

To implement the Kuhn-Tucker (KKT) conditions using the Lagrangian Multiplier method in Python to solve constrained optimization problems with inequality constraints.

PROCEDURE:

1. Define the objective function and inequality constraints symbolically using sympy.
2. Formulate the Lagrangian $L = f \pm \lambda \cdot g$ for each constraint.
3. Derive KKT stationarity conditions by differentiating L with respect to each variable.
4. Add complementary slackness conditions $\lambda \cdot g = 0$.
5. Solve the KKT system and filter feasible solutions satisfying $\lambda \geq 0$ and $g(x) \leq 0$.

PROBLEM 1: Minimize $f(x,y) = x^2 + y^2$ s.t. $x + 2y \leq 4$, $x,y \geq 0$

CODE

```
import sympy as sp
x, y = sp.symbols('x y', real=True, nonneg=True)
lam = sp.symbols('lambda', real=True)
f = x**2 + y**2
g = x + 2*y - 4 # g <= 0 form
L = f + lam * g
dLdx = sp.diff(L, x); dLdy = sp.diff(L, y)
kkt = [dLdx, dLdy, lam * g]
sols = sp.solve(kkt, [x, y, lam], dict=True)
print("KKT Solutions:")
for sol in sols:
    gv = g.subs(sol); lamv = sol.get(lam, 0)
    if gv <= 0 and lamv >= 0:
        fv = f.subs(sol)
        print(" x={}, y={}, lambda={}".format(sol[x], sol[y], lamv))
        print(" Minimum f(x,y) =", fv)
```

OUTPUT

```
KKT Solutions:
x=0, y=0, lambda=0
Minimum f(x,y) = 0
```

PROBLEM 2: Maximize $f(x,y) = 4x + 3y$ s.t. $2x + y \leq 10$, $x + 3y \leq 12$, $x,y \geq 0$

CODE

```

x, y = sp.symbols('x y', real=True, nonneg=True)
l1, l2 = sp.symbols('l1 l2', real=True)
f = -(4*x + 3*y)
g1 = 2*x + y - 10; g2 = x + 3*y - 12
L = f + l1*g1 + l2*g2
kkt = [sp.diff(L,x), sp.diff(L,y), l1*g1, l2*g2]
sols = sp.solve(kkt, [x, y, l1, l2], dict=True)
for sol in sols:
    if (g1.subs(sol)<=0 and sol.get(l1,0)>=0 and
        g2.subs(sol)<=0 and sol.get(l2,0)>=0):
        obj = (4*x + 3*y).subs(sol)
        print(" x={}, y={}, Z={}".format(sol[x], sol[y], obj))

```

OUTPUT

```

Feasible KKT Solutions:
x=9/5, y=32/5, Z=132/5

```

PROBLEM 3: Minimize $f(x,y,z) = x^2+y^2+z^2$ s.t. $x+y+z \leq 3$, $x,y,z \geq 0$

CODE

```

x,y,z = sp.symbols('x y z', real=True, nonneg=True)
lam = sp.symbols('lambda', real=True)
f = x**2 + y**2 + z**2
g = x + y + z - 3
L = f + lam * g
kkt = [sp.diff(L,x), sp.diff(L,y), sp.diff(L,z), lam*g]
sols = sp.solve(kkt, [x, y, z, lam], dict=True)
print("Feasible KKT Solutions:")
for sol in sols:
    gv = g.subs(sol); lamv = sol.get(lam, 0)
    if gv <= 0 and lamv >= 0:
        fv = f.subs(sol)
        print(" x={}, y={}, z={}".format(sol.get(x,0), sol.get(y,0), sol.get(z,0)))
        print(" Minimum f =", fv)
        print(" lambda =", lamv)

```

OUTPUT

```

Feasible KKT Solutions:
x=0, y=0, z=0
Minimum f = 0
lambda = 0

```

RESULT:

Thus, the Kuhn-Tucker Conditions using Lagrangian Multiplier Method has been implemented and executed successfully.